

Agenda

- Count total no of set bits
- finding good days
- Unique element
- Subarray of OR equals 0
- " " " " 1.

Q Given a no. n , count total no of set bits of n .

$n = 12 \longrightarrow 1100 \longrightarrow \text{Ans} = 2.$

$n = 15 \longrightarrow 1111 \longrightarrow \text{Ans} = 4.$

Approach Left shift + And.

Right shift

```
boolean checkSetBit(int n, int k) {  
    if (n & (1 << i) == 0) {  
        return false;  
    }  
    return true;  
}  
  
int countSetBits(n) {  
    count = 0;  
    for (i = 0; i < 32; i++) {  
        if (checkSetBit(n, i)) {  
            count += 1;  
        }  
    }  
    return count;  
}
```

TC: $O(32) \sim O(1) \sim O(\log_2 n)$
SC: $O(1)$

Qu2 Alex $\rightarrow$ cat $\rightarrow$ Boomer. He decides to put his cat to test for eternity.

starts on day 1 with one stash of food unit.
every next day stash doubles

If boomer is well behaved during a particular day, only then she receives food worth equal to stash produced on that day.

Boomer receives a net worth of A units of food.
What is the no of days she received the stash?
[ripping]

Ex: $A = 5$

single day = 5

$$5 \Rightarrow 2^2 + 2^0$$

1 day (4) 1 day (1) = 2 days

1 0 1
↑ ↑ 2 Ans

$A = 13$

$$13 \rightarrow 2^3 + 2^2 + 2^0$$

8 4 1

Day 1 $\rightarrow$ 8
2 $\rightarrow$ 4
3 $\rightarrow$ 1

3 days

1 1 0 1
↑ ↑ ↑ Ans = 3

$A = 8$

Day 1 $\rightarrow$ 8

1 days

1 0 0 0
↑ 1 Ans

At any particular day ———

Day 1 ——— $2^0 = 1$ power of 2

Day 2 ——— $2^1 = 2$ power of 2

3 ——— $2^2 = 4$

Day 1 ——— 1 stash

Day 2 ——— 2 "

3 ——— 4

4 ——— 8

5 ——— 16

6 ——— 32

$A = 73$

73 chocolates = 1 X

73:

$$64 + 8 + 2 + 1$$

$2^6 + 2^3 + 2^1 + 2^0 \rightarrow$ 4 days

11
↑
3
1

Pseudocode

```
int calculateDays(n) {  
    return countSetBits(n);  
}
```

TC: $O(1)$ || $O(\log_2 n)$
SC: $O(1)$

Qu $A[]$, all elements occurs thrice, except for one.
find unique element.

Amazon
Microsoft
Cred

4	5	5	4	1	6	6	4	5	6
---	---	---	---	---	---	---	---	---	---

output: 1 Ans

Brute force:

```

for(i=0 ————— n-1) {
    cnt=0;
    for(j=0 ————— j=n-1) {
        if (A[i] == A[j]) {
            cnt++;
        }
    }
    if (cnt==1) {
        return A[i];
    }
}

TC:  $O(n^2)$ 
SC:  $O(1)$ 

```

Approach 2

sort ————— $n \log_2 n$
 $O(1)$

Approach 3

Map ————— $O(n)$
 $O(n)$

Expected
Approach: $O(n)$
 $O(1)$

Approach 4: XOR ~~X~~

4	5	5	4	1	6	6	4	5	6
---	---	---	---	---	---	---	---	---	---

$$\begin{array}{r}
 4^4 5^4 6^4 1^1 \\
 \underline{4^4 4^4 4^4} \quad \underline{5^4 5^4 5^4} \quad \underline{6^4 6^4 6^4} \quad 1^1 \\
 0 \quad \quad \quad 0 \quad \quad \quad 0 \quad \quad \quad 1 \\
 \hline
 4^4 5^4 6^4 1^1
 \end{array}$$

Approach All no appear odd no of times.

5	7	5	9	7	11	11	7	5	11
---	---	---	---	---	----	----	---	---	----

corresponding $2^i \leftarrow$ 8 4 2 1

indices $\leftarrow$ 3 2 1 0

5	0	1	0	1
7	0	1	1	1
5	0	1	0	1
9	1	0	0	1
7	0	1	1	1
11	1	0	1	1
11	1	0	1	1
7	0	1	1	1
5	0	1	0	1
11	1	0	1	1

(1) $\left\{ \begin{matrix} 6 \text{ 0's} \\ 4 \text{ 1's} \end{matrix} \right\} \quad 0 \left(\begin{matrix} 4 \text{ 0's} \\ 6 \text{ 1's} \end{matrix} \right) \quad 0 \left\{ \begin{matrix} 4 \text{ 0's} \\ 6 \text{ 1's} \end{matrix} \right\} \quad 1 \left\{ \begin{matrix} 0 \text{ 0's} \\ 10 \text{ 1's} \end{matrix} \right\}$

Remove 9

$\begin{matrix} 6 \text{ 0's} & 30's & 30's & 0 \text{ 0's} \\ 3 \text{ 1's} & 6 \text{ 1's} & 6 \text{ 1's} & 9 \text{ 1's} \end{matrix}$

$\boxed{1 \ 0 \ 0 \ 1} \Rightarrow \underline{9 \text{ Ans}}$

5 times $\Rightarrow$ Multiple of 5

7 times $\Rightarrow$ " " 7

...

Qn A[], all elements occurs ~~thrice~~, except for one.
 find unique element: ~~twice~~ | ~~four~~

2	4	2	1	6	1	4	9	8	8	11	9	6
---	---	---	---	---	---	---	---	---	---	----	---	---

$$2 \wedge 4 \wedge 2 \wedge 1 \wedge 6 \wedge 1 \wedge 4 \wedge 9 \wedge 8 \wedge 8 \wedge 11 \wedge 9$$

$$\underline{2 \wedge 2} \wedge \underline{4 \wedge 4} \wedge \underline{1 \wedge 1} \wedge \underline{6 \wedge 6} \wedge \underline{8 \wedge 8} \wedge \underline{9 \wedge 9} \wedge 11$$

$$0 \wedge 11 = 11 \text{ Ans}$$

All no appear even times ——— xor work

Pseudocode

```
ans = 0;
for(i = 0 ——— 31) {
    cnt = 0;
    for(j = 0 ——— n-1) {
        if (checksetBit(A[j], i)) {
            cnt++;
        }
    }
    if (cnt % 3 != 0) {
        ans += Math.pow(2, i);
        (OR)
        ans = ans | 1 << i;
    }
}
return ans;
```

[4, 3, 0, 4, 3, 4, 3]

4	—	1	0	0
3	—	0	1	1
0	—	0	0	0
4	—	1	0	0
3	—	0	1	1
4	—	1	0	0
3	—	0	1	1
		40's	40's	40's
		[0]	[0]	[0]

TC: $O(32 * n) \approx O(n)$

SC: $O(1)$

Break : 10:15 - 10:25

Qn $A[] \rightarrow$ 0s and 1s. { Google type }

Calculate no of subarrays for which the bitwise or of all elements in subarray is 0.

1	0	0	1
---	---	---	---

$\rightarrow$ [1] ——— 1
 [1 0] ——— 1
 [1 0 0] ——— 1
 [1 0 0 1] ——— 1
 [0] ——— 0
 [0 0] ——— 0
 [0 0 1] ——— 1
 [0] ——— 0
 [0 1] ——— 1
 [1] ——— 1

Ans = 3

Subarray ——— bitwise OR = 0
 It means all el of that array are 0.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	1	0	0	0	0	1	1	0	0	1	0	0	0

$\downarrow \downarrow \downarrow$
 $\frac{3(3+1)}{2}$
 $\frac{3*4}{2} = 6$
 $\downarrow$ [0]
 $\downarrow$ [0 0]
 $\downarrow$ [0 0 0]
 $\downarrow$ [0]
 $\downarrow$ [0 0]
 $\downarrow$ [0]
 $\downarrow$ [0]

$\downarrow \downarrow \downarrow \downarrow$
 $\frac{4(4+1)}{2}$
 $\frac{4*5}{2} = 10$

$\downarrow \downarrow$
 $\frac{2(2+1)}{2}$
 $\frac{2(3)}{2}$
 $\frac{2}{3}$

special case

consecutive zeros = ~~3~~ ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~0~~ ~~1~~ ~~2~~ ~~0~~ 3

$$\begin{aligned}
 \text{ans} &\Rightarrow \boxed{6 + 10 + 3} = 19 + \left[\frac{3*(3+1)}{2} \right] \\
 &= 19 + 6 = \boxed{25} \text{ Ans}
 \end{aligned}$$

Pseudocode

int countSubarrayWithBitORas0 (A[]) {

ans = 0

zeros = 0;

for (i = 0 _____ n-1)

if (A[i] == 0) {

zeros++;

} else {

ans += $\left[\frac{\text{zeros} * (\text{zeros} + 1)}{2} \right]$

zeros = 0; ;

}

ans += $\left[\frac{\text{zeros} * (\text{zeros} + 1)}{2} \right]$

return ans;

TC: $O(n)$

SC: $O(1)$

Que

$A[] \rightarrow$ 0s and 1s.

Calculate no of subarrays for which the bitwise or of all elements in subarray is 1.

Ans: total no of subarrays — count subarray with bit OR as 0 (A);
↓
zeroAns

Ans: $\frac{n(n+1)}{2} - \text{zeroAns}$ Ans

TC: $O(n)$

sc: $O(1)$

Thankyou 😊